## Osman Gültekin


**Contact Information**

Middle East Technical University (METU)
Department of Civil Engineering
Structural Engineering Division, K2 Building, room 306
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**Profiles**

- Google Scholar : CB920AAAAJ (url)
- Scopus : 57189073759 (url)
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- Web of Science : AAD-5715-2019 (url)

**Research Interests**

- Solid Mechanics and Constitutive Theory
- Computational Mechanics
- Biomechanics
- Computational Modeling of Multi–physics Problems
- Machine Learning applications in Mechanics

**Education**

- **Ph.D.**, Institute of Biomechanics, (11/2018)
  Graz University of Technology, Austria

  Thesis: Computational Inelasticity of Fibrous Biological Tissues with a Focus on Viscoelasticity, Damage and Rupture
  Supervisor: Prof. Dr. Gerhard A. Holzapfel
  Description: Developing multi-field (mechano-fracture) continuum mechanical frameworks and computational algorithms, e.g. user material and element subroutines along with pre- and post-processing tools for FEM, predicting the finite inelastic response, such as anisotropic viscoelasticity in human myocardium, damage and fracture occuring in human arterial walls and aorta in health and disease, e.g., aortic dissection, atherosclerosis, aneurysms.

- **M.Sc.**, Computational Mechanics of Materials and Structures, (01/2014)
  University of Stuttgart, Germany

  Thesis: A Phase-field Approach to the Fracture of Anisotropic Medium
  Supervisor: Prof. Dr. Christian Miehe , Dr. Hüsnü Dal
  Description: Developing continuum mechanical frameworks and computational algorithms, e.g., user material and element subroutines, for the finite elastic and fracture responses of soft biological tissues and generating finite element meshes for an atherosclerotic arterial wall

- **B.Sc.**, Civil Engineering, (06/2011)
  Middle East Technical University, Ankara, Turkey
  Specialization: Structural Mechanics

**Awards and Honours**

- Invited speaker in the ECCOMAS Young Investigator Conference 2019, September 1-6, 2019, Krakow, Poland
  free participation & accommodation (not taken)
- Winner of the CEACM nomination for the ECCOMAS Award for the Two Best PhD Theses on Computational Methods in Applied Sciences and Engineering in 2018
- Graz University of Technology, Ph.D.
  Graduation with distinction: Dissertation 1.0/1.0 & Defense 1.0/1.0

- University of Stuttgart, M.Sc.
  Graduation with high honor CumGPA 1.5/1.0 [1] & Thesis: 1.0/1.0
- Middle East Technical University, B.Sc.
  Graduation with high honor CumGPA 3.72/4.0 [2]
- Granted DAAD-TEV Master's study scholarship offered by the German Academic Exchange Service (DAAD) in cooperation with the Turkish Education Foundation (TEV) (2011–2013)

**Academic Experience**

**2013 – present**

- Middle East Technical University (04/2026 – present)
  *Asst. Prof. Dr.*
- Dresden University of Technology (12/2023 – 03/2025)
  *Postdoctoral Research Fellow*
- The University of Texas at Austin (07/2022 – 07/2023)
  *Research Fellow*
- Middle East Technical University (01/2019 – 05/2022)
  *Postdoctoral Research Fellow*
- Graz University of Technology (02/2014–11/2018)
  *Research Associate*
- University of Stuttgart (01/2013–01/2014)
  *Student and administrative asisstant* for international M.Sc. program Computational Mechanics of Materials and Structures

**Teaching Experience**

- Middle East Technical University, *Lecturer, Teaching Asisstant* (01/2019 – 05/2022)
  *BME 501 - Introduction to Biomedical Engineering-Biomechanics*
  **Spring: 2020, 2021**
  *ME 547 - Introduction to Continuum Mechanics* (Jointly with Prof. Dr. Hüsnü Dal )
  **Fall: 2019, 2020**
  *ME 210 - Applied Mathematics for Mechanical Engineers* (Jointly with Prof. Dr. Hüsnü Dal )
  **Fall: 2019**
- Graz University of Technology, *Lecturer* (02/2014 – 11/2018)
  *717.327 - Biophysical Modeling* (Jointly with Andreas Petrovic, M.Sc.)
  **Spring: 2017, 2018**

**Theses Supervised**

- Traineeship
  - Erasmus traineeship program "Identification of the fiber-sheet orientation for the truncated biventricular heart model based on realistic data", Dominykas Gudavicius, 2015 [3]
- Bachelor's theses
  - B.Sc. Thesis "One-dimensional sensitivity analysis of different failure criteria in the context of crack phase-field modeling ", Alfonso Doménech Réal, 2017[3]
- Master's theses
  - M.Sc. Thesis "Crack phase-field modeling of the atherosclerotic rupture", Jakob Eckmann, 2020 (Co-supervisors: Prof. Dr. Gerhard A. Holzapfel) [3]
  - M.Sc. Thesis "A crack phase-field approach to model aortic dissections", Sandra P. Hager, 2018 (Co-supervisors: Prof. Dr. Gerhard A. Holzapfel) [3]

[1] 1.0 is the highest grade
[2] 4.0 is the highest grade
[3] at Institute of Biomechanics

**Funding Acquired**

- "*Fast Prediction of Transmural Mechanics in Living Heart via Physics-informed Graph Neural Networks (TransMechHeart-PIGNN)*" (under review), 3-year Deutsche Forschungsgemeinschaft (DFG) project, €621,000
(together with Prof. Dr. M. Kaliske, Dr. Baris Cansız)
- "*Experimental and Numerical Analysis of Thermomechanical Fracture Processes in Polymers Accounting for Ductile to Brittle Transition (THERMEFRACPOL)*" (2020), 2-year TUBITAK BIDEB 2236 Co-Funded Brain Circulation Scheme 2 project (European Commission Horizon 2020 Marie Sklodowska-Curie Actions Cofund program), €287,040

**Skills**

- Languages: Turkish (native), English (YDS: 98.75 [2025], IELTS: 7.5 [2019]), and German (B1 Level)
- Programming Languages: Fortran, Python
- FE packages: FEAP, Abaqus®, FEBio
- Scientific Computing Programs: Matlab® , Maple® , Tecplot® , Trelis(Cubit)®, Paraview, Gnuplot, Shell Script, Latex
- Operating Systems: Unix/Linux and Windows

**Professional Activities**

- Organization
  - Organization of the Symposium "Tecplot User Day: Postprocessing of Simulation Data", on May 27$^{\text{th}}$ 2019, Department of Mechanical Engineering, METU, Ankara
- Reviewer for
  - Computer Methods in Biomechanics and Biomedical Engineering
  - Biomechanics and Modeling in Mechanobiology
  - European Journal of Mechanics / A Solids

**Publications**

43 publications in total, thereof:

- 12 peer-reviewed journal publications (international)
- 1 peer-reviewed journal publication (national)
- 3 book chapters (international)
- 3 conference proceeedings (full papers)
- 22 conference abstracts, oral presentations, invited talks
- 2 technical reports

**Journal Publications (International)**

J12 V. Dubey, C. E. Haese, **O. Gültekin**, D. Dalton, M. K. Rausch, J. Fuhg [2026]: *Graph Neural Network Surrogates for Contacting Deformable Bodies with Necessary and Sufficient Contact Detection* Computer Methods in Applied Mechanics and Engineering 448, Part A, 118413. (url)
Category Quartile (WOS) : **Q1**

J11 **O. Gültekin**, A. Moeineddin, B. Cansız, K. Sveric, A. Linke, M. Kaliske [2025]: *A physics-informed neural network model for the anisotropic hyperelasticity of the human passive myocardium* International Journal for Numerical Methods in Engineering 126, no. 13: e70067. (url)
Category Quartile (WOS) : **Q1**

J10 **O. Gültekin**, M. J. Lohr, G. N. Bechtel, M. K. Rausch [2024]: *"What makes blood clots break off?" A back-of-the-envelope computation toward explaining clot embolization* Cardiovascular Engineering and Technology 15, 584–593. (url)
Category Quartile (WOS) : **Q3**

J9 H. Dal, **O. Gültekin**, S. Başdemir, A. K. Açan [2022]: *Ductile-brittle failure of amorphous glassy polymers: A phase-field approach* Computer Methods in Applied Mechanics and Engineering 401, 115639. (url)
Category Quartile (WOS) : **Q1**

J8 H. Dal, **O. Gültekin**, K. Açıkgöz [2020]: *An extended eight-chain model for hyperelastic and finite viscoelastic response of rubberlike materials: Theory, experiments and numerical aspects* Journal of the Mechanics and Physics of Solids 145, 104159. (url)
Category Quartile (WOS) : **Q1**

J7 **O. Gültekin**, B. Rodoplu, H. Dal [2020]: *A quasi-incompressible and quasi-inextensible finite element analysis of fibrous soft biological tissues* Biomechanics and Modeling in Mechanobiology 19, 2357–2373. (url)
Category Quartile (WOS) : **Q3**

J6 F. A. Denli, **O. Gültekin**, G. A. Holzapfel, H. Dal [2020]: *A phase-field model for fracture of unidirectional fiber-reinforced polymer matrix composites* Computational Mechanics 65, 1149–1166.(url)
Category Quartile (WOS) : **Q1**

J5 **O. Gültekin**, S. P. Hager, H. Dal, G. A. Holzapfel [2019]: *Computational modeling of progressive damage and rupture in fibrous biological tissues: application to aortic dissection* Biomechanics and Modeling in Mechanobiology 18, 1607–1628. (url)
Category Quartile (WOS) : **Q2**

J4 **O. Gültekin**, H. Dal, G. A. Holzapfel [2019]: *On the quasi-incompressible finite element analysis of anisotropic hyperelastic materials* Computational Mechanics 63, 443–453. (url)
Category Quartile (WOS) : **Q1**

J3 **O. Gültekin**, H. Dal, G. A. Holzapfel [2018]: *Numerical Aspects of Anisotropic Failure in Soft Biological Tissues Favor Energy-based Criteria: A Rate-dependent Anisotropic Crack Phase-field Model*. Computer Methods in Applied Mechanics and Engineering 331, 23–52. (url)
Category Quartile (WOS) : **Q1**

J2 **O. Gültekin**, H. Dal, G. A. Holzapfel [2016]: *A phase-field approach to model fracture of arterial walls: Theory and finite element analysis*. Computer Methods in Applied Mechanics and Engineering 312, 542–566. (url)
Category Quartile (WOS) : **Q1**

J1 **O. Gültekin**, G. Sommer, G. A. Holzapfel [2016]: *An orthotropic viscoelastic model for the passive myocardium: continuum basis and numerical treatment*. Computer Methods in Biomechanics and Biomedical Engineering 19, 1647–1664. (url)
Category Quartile (WOS) : **Q3**

**Journal Publications (National)**

J1 **O. Gültekin**:(in press) *Searching for the Poynting Effect in Constitutive Models of Arterial Walls*. Journal of Polytechnic (Politeknik Dergisi)
Category Quartile : **TR-Dizin**

**Book Chapters (International)**

B3 **O. Gültekin**, H. Dal [2022]: *Computational Finite Strain Orthotropic Viscoelasticity of Human Passive Myocardium*. In: G. Sommer, K. Li, D. C. Haspinger, R. W. Ogden (eds) Solid (Bio)mechanics: Challenges of the Next Decade. Studies in Mechanobiology, Tissue Engineering and Biomaterials, vol 24. Springer, Cham. (url)

B2 **O. Gültekin**, G. A. Holzapfel [2022]: *Anisotropic Failure Criteria in Relation to Crack Phase-Field Modeling at Finite Strains*. In: F. Aldakheel, B. Hudobivnik, M. Soleimani, H. Wessels, C. Weißenfels, M. Marino (eds) Current Trends and Open Problems in Computational Mechanics. Springer, Cham. (url)

B1 **O. Gültekin**, G. A. Holzapfel [2018]: *A brief review on computational modeling of rupture in soft biological tissues. Computational methods in applied sciences*. In E. Oñate, D. Peric, E. de Souza Neto, and M. Chiumenti (Editors), Advances in Computational

Plasticity. A Book in Honour of D. Roger J. Owen, volume 46(6), pp. 113–144. Springer Nature. (url)

**Conference Proceedings (Full Papers)**

CP3 S. Başdemir, **O. Gültekin**, H. Dal [2023]: *Coupled thermo-viscoplastic fracture model for ductile-brittle failure of amorphous glassy polymers with phase-field approach.* Proceedings of Applied Mathematics and Mechanics (PAMM), 22:1 e202200280. (url)

CP2 **O. Gültekin**, H. Dal, G. A. Holzapfel [2017]: *Crack Phase-Field Modeling of Anisotropic Rupture in Fibrous Soft Tissues.* In E. Oñate, D. Peric, E. de Souza Neto, and M. Chiumenti (Editors) Proceedings of the XIV International Conference on Computational Plasticity - Fundamentals and Applications COMPLAS XIV, 5-7 September, pp. 139–150.(url)

CP1 H. Dal, **O. Gültekin**, F. A. Denli and G. A. Holzapfel [2017]: *Phase-field models for the failure of anisotropic continua.* Proceedings of Applied Mathematics and Mechanics (PAMM), 17:91–94. (url)

**Oral Presentations - (Extended) Abstracts**

G. A. Holzapfel, M. Rolf, **O. Gültekin** [2025]: *Crack phase-field modeling to predict the progression of aortic dissection* EUROMECH Colloquium 655, 25 – 27 August 2025, Graz, Austria

V. Dubey, C. E. Haese, J. Fuhg, M. K. Rausch, **O. Gültekin** [2024]: *Graph Neural Networks for Contact Between Deformable Bodies.* TACCSTER 2024 Texas Advance Computing Center Symposium for Texas Researchers, September 23–25, 2024, Austin, Texas, USA. (url)

**O. Gültekin**, B. Cansız, A. Moeineddin, M. Kaliske [2024]: *Surrogate modeling of finite deformation hyperelasticity of human myocardial tissue.* VPH 2024 Virtual Physiological Human Conference, September 4–6, 2024, Stuttgart, Germany

**O. Gültekin** [2022]: *Phase–field modeling of soft tissue fracture.* Seminar at the Oden Institute for Computational Engineering & Sciences, December 8, 2022, Austin, Texas, USA. **(Invited Talk)**

S. Başdemir, **O. Gültekin**, H. Dal [2022]: *Coupled thermo-viscoplastic fracture model for ductile-brittle failure of amorphous glassy polymers with phase-field approach.* GAMM 2022 92nd Annual Meeting of the International Association of Applied Mathematics and Mechanics, August 15–19, 2022, Aachen, Germany

**O. Gültekin**, H. Dal, F. Welschinger [2021]: *Microstructure based modeling of shear yielding and crazing induced fracture in amorphous polymers.* COMPLAS 2021 16th International Conference on Computational Plasticity-Fundamentals and Applications, September 7-9, 2021, Barcelona, Spain.

**O. Gültekin**, F. Welschinger, H. Dal [2021]: *A unified microstructure-based approach for ductile and brittle fracture of amorphous glassy polymers.* ICTAM 2020+1 25th International Congress of Theoretical and Applied Mechanics of IUTAM, August 23-28, 2021, Milano, Italy.

**O. Gültekin** [2019]: *Biomechanics from its Histological and Pathophysiological Aspects to Experimental and Numerical Investigations.* Colloquium at Institute of Applied Mathematics, Middle East Technical University, November 19, 2019, Ankara, Turkey. **(Invited Talk)**

**O. Gültekin**, B. Rodoplu, and H. Dal [2019]: *A quasi-incompressible and quasi-inextensible finite element implementation of fibrous soft biological tissues.* BEYOND 2019 Computational Science and Engineering Conference, September 9-11, 2019, Ankara, Turkey

**O. Gültekin**, G. A. Holzapfel, and H. Dal [2019]: *Phase-field approach to model fracture in human aorta.* IWPDF 2019 1st International Workshop on Plasticity, Damage and Fracture of Engineering Materials, August 22-23, 2019, Ankara, Turkey.

F. A. Denli, **O. Gültekin**, and H. Dal [2019]: *A phase-field approach to viscoelastic fracture in rubbery polymers.* IWPDF 2019 1st International Workshop on Plasticity, Damage and Fracture of Engineering Materials, August 22-23, 2019, Ankara, Turkey.

G. A. Holzapfel, **O. Gültekin**, S. P. Hager and H. Dal [2019]: *A crack phase-field model to analyze aortic dissections.* COMPLAS 2019 15th International Conference on Computational Plasticity - Fundamentals and Applications, September 3-5, 2019, Barcelona, Spain.

S. P. Hager, **O. Gültekin** and G. A. Holzapfel [2019]: *A crack phase-field approach to model aortic dissections.* ESB 2019 25th Congress of the European Society of Biomechanics, July 7-10, 2019, Vienna, Austria.

J. Eckmann, **O. Gültekin** and G.A. Holzapfel [2019]: *Stability of crack growth in anisotropic materials at finite strains.* GAMM 2019 90th Annual Meeting of the International Association of Applied Mathematics and Mechanics, February 18–22, 2019, Vienna, Austria.

**O. Gültekin**, H. Dal, and G. A. Holzapfel [2018]: *Numerical Modeling of Rupture in Human Arterial Walls.* WCB 2018 8th World Congress of Biomechanics, July 8-12, 2018, Dublin, Ireland.

**O. Gültekin**, H. Dal, and G. A. Holzapfel [2018]: *A Phase-field Approach to Model Aortic Dissections.* ESMC 2018 10th European Solid Mechanics Conference, July 2-6, 2018, Bologna, Italy.

G. A. Holzapfel, **O. Gültekin** and G. Sommer [2018]: *Viscoelastic modeling of the passive myocardium.* ESMC 2018 10th European Solid Mechanics Conference, July 2-6, 2018, Bologna, Italy.

G. A. Holzapfel, J. A. Niestrawska and **O. Gültekin** [2017]: *Modeling the microstructure and mechanics of human artery walls.* ICBT 2017 3rd ECCOMAS Thematic Conference on Biomedical Technology, November 6-8, 2017, Hannover, Germany.

**O. Gültekin**, H. Dal, and G. A. Holzapfel [2017]: *Phase-field Modeling of Rupture in Soft Biological Tissues.* COMPLAS 2017 XIV International Conference on Computational Plasticity - Fundamentals and Applications, September 5-7, 2017, Barcelona, Spain.

G. A. Holzapfel, J. A. Niestrawska, **O. Gültekin**, and R. W. Ogden [2017]: *The importance of considering the microstructure of soft biological tissues: A general fiber dispersion model with related analysis.* Maths from the body, May 29-31, 2017, Brescia, Italy.

H. Dal, **O. Gültekin**, F. Denli and G. A. Holzapfel [2017]: *Phase field models for the failure of anisotropic continuum.* GAMM 2017 88th Annual Meeting of the International Association of Applied Mathematics and Mechanics, March 6-10, 2017, Weimar, Germany.

G. A. Holzapfel, **O. Gültekin**, S. Sherifova, G. Sommer and B. E. Griffith [2016]: *Experimental and computational analysis of aortic dissection.* ICTAM 2016 24th International Congress of Theoretical and Applied Mechanics of IUTAM, August 21-26, 2016, Montréal, Québec, Canada.

**O. Gültekin**, H. Dal, and G. A. Holzapfel [2016]: *A Phase-field Approach to Model Fracture of Arterial Walls.* ECCOMAS 2016 European Congress on Computational Methods in Applied Sciences and Engineering, June 5-10, 2016, Crete, Greece

**O. Gültekin**, G. Sommer , and G. A. Holzapfel [2015]: *An Orthotropic Viscoelastic Formulation for the Passive Myocardium: Continuum Basis and Numerical Treatment.* ESMC 2015 9th European Solid Mechanics Conference, July 6–10, 2015, Madrid, Spain.

**Technical Reports**

**O. Gültekin** [2018]: *Computational Inelasticity of Fibrous Biological Tissues with a Focus on Viscoelasticity, Damage and Rupture*, Ph.D. Thesis, Institute of Biomechanics, Graz University of Technology, 207 pages.

**O. Gültekin** [2014]: *A Phase-Field Approach to the Fracture of Anisotropic Medium*, M.Sc. Thesis, Computational Mechanics of Materials and Structures, University of Stuttgart, 73 pages.